

Boost Performance with Real-World Examples and SQL Commands

 systemdesignschool.io/fundamentals/database-partition-codelab



Database Partition Tutorial

Let's get our hands dirty with a real example using MySQL/MariaDB. Suppose we have a comment system with a table of comments. We can partition the table by the year the comment was created.

```
CREATE TABLE comments ( comment_id INT NOT NULL, page_id INT NOT NULL, user_id INT NOT NULL, content TEXT NOT NULL, created_time DATETIME NOT NULL ) PARTITION BY RANGE (year(created_time)) ( PARTITION pold VALUES LESS THAN (2019), PARTITION p19 VALUES LESS THAN (2020), PARTITION p20 VALUES LESS THAN (2021), PARTITION p21 VALUES LESS THAN (2022), PARTITION p22 VALUES LESS THAN (2023) );
```

Then insert two comments: the first comment is from 2009, and the second comment is from 2021.

```
INSERT INTO `comments` ( `comment_id`, `page_id`, `user_id`, `content`, `created_time` ) VALUES ( '1', '1', '1', 'The Times 03/Jan/2009 Chancellor on brink of second bailout for banks', '2009-01-03 18:15:05' ), ( '2', '2', '2', 'Hello algo.monster', '2021-10-11 18:45:02' )
```

To check the result, we can use the following queries.

```
-- select all comments from all partitions SELECT * FROM comments; -- select only in the "pold" partition SELECT * FROM comments PARTITION (pold); -- select only in the "p21" partition SELECT * FROM comments PARTITION (p21);
```

The first query should print both comments. The second query shows the comment from 2009, and the third query shows the comment from 2021. You can try the code here:

<http://sqlfiddle.com/#!9/fccec/3>.